MIL-S-23566A(SHIPS)
3 August 1964
SUPERSEDING
MIL-S-23566(SHIPS)
8 February 1963
(See 6.2)

MILITARY SPECIFICATION

SINK, SANITIZING (SUBMARINE SERVICE)

1. SCOPE

1.1 Scope.- This specification covers electrically heated sinks for use in sanitizing mess gear (dishes) aboard submarines.

1.2 Classification.- Sinks shall be of the following sizes, as specified (see 6.1):

Size I - 14 x 18 x 20 inches deep
Size II - 16 x 20 x 20 inches deep

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)
Standard 45 - Recommended Practice for Electric Installations on Shipboard.

(Application for copies should be addressed to the Institute of Electrical and Electronics Engineers, Box A, Lenox Hill Station, New York 21, N. Y.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Preproduction.- Prior to beginning production, a sample consisting of one sink of each size specified in the contract or order shall be tested as specified in 4.2.

3.2 Material.- Thickness of materials shall be consistent with good design and suitable for the purpose intended.

3.2.1 Sink body, cover, sink heater body, and insulation cover shall be made of corrosion-resisting steel properly normalized, where required, to prevent stress corrosion.

3.2.2 Insulating material shall be at least equivalent to bonded fibrous glass felt with a nominal density of 1.0 pound per cubic foot and an average fiber diameter of 0.00025 inch.

3.2.3 All other materials shall be resistant to corrosion or suitably and durably protected to resist corrosion.

3.3 Design.- The unit shall essentially consist of a sink with an electrically heated, thermostatically controlled sink heater, securely attached to the under side of the sink, having controls and safety devices, as hereinafter specified, to heat and retain water within the sink at a temperature of 180 to 190 degrees F. for sterilizing mess gear (dishes). The unit shall function by the sink water circulating through the heater reservoir (where it is heated by electric immersion type heaters) and back to the sink by convection currents.

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3.3.1 Sink requirements.- The main body of the units shall be a size to pass through a hatch having a maximum inside diameter of 25 inches. To accomplish this requirement and retain maximum possible inside sink width and length dimensions of the size II unit, the sink rim and insulating jacket shall be designed for attachment to the unit after being installed within the submarine.

3.3.1.1 Dimensions.- Heights of units, including support base, shall be 36 inches with provision for leveling and adjusting to a height of not less than 39 inches.

3.3.1.1.1 The inside dimensions of the size I sink shall be 14 inches wide (front to back) by 18 inches long by 20 inches deep.

3.3.1.1.2 The inside width (front to back) and length of the size II sink shall be as near 16 inches by 20 inches as the limitations under 3.3.1 permit. The size II sink shall have an internal depth of 20 inches.

3.3.1.2 Corner construction.- Internal corners of sinks shall have an approximate 1-inch radius. No laps, crevices or thermostatic elements are permitted within the sink proper.

3.3.1.3 Overflow.- Each sink shall have an integral open overflow arrangement to maintain a water level of 16 inches within the sink. The overflow shall be fitted with a removable corrosion-resistant metal strainer.

3.3.1.4 A 1-1/2 inch diameter plug cock shall be secured to the underside of the sink below the waste outlet. Handle for operating the plug cock shall be conveniently located. Waste outlet shall be fitted with recessed readily removable (without tools) basket type strainer. Strainer openings shall be equal to area of 1-1/2 inch pipe.

3.3.2 Heater requirements.- The sink heater shall essentially consist of a water heating reservoir, a waste collecting reservoir, controls, safety devices, and means of attaching to the sink. The unit shall heat and maintain water in the sink at a temperature of 180°F. to 190°F.

3.3.2.1 Dimensions.- Length of heater shall not exceed 18 inches, width shall not exceed 8 inches and height (exclusive of plumbing fittings) shall not exceed 10-1/4 inches.

3.3.2.2 Connections.- Intake and outlet plumbing connections shall be provided for attaching the heater to the sink. Overall height of connections shall not exceed 2-1/2 inches when installation to sink is completed. Connections shall be along center top of the unit, approximately 5 inches from each end.

3.3.2.3 Waste reservoir.- A waste collecting reservoir shall be provided to remove and trap waste particles from the main stream of the circulating water.

3.3.2.3.1 Waste outlet.- The waste reservoir shall be fitted with a quick opening discharge valve for purpose of draining the waste reservoir without draining the heater reservoir. The valve operating handle shall be readily accessible to the operator when facing the front of the sink.

3.3.3 Electrical requirements.- All electrical equipment shall conform to the recommended practices for electrical installation on shipboard as required by IEEE Publication 45. All electrical equipment shall be designed for operation on an ungrounded electrical system, but shall operate satisfactorily in accordance with this specification without hazard to personnel or equipment in event of an accidental ground on any conductor.

3.3.3.1 Control system.- The control system shall consist of the contactors, relays, thermostats, and indicators necessary to provide temperature control and indication of operation. On the 440 volt (v.), alternating current (a.c.), sink heater, the control system shall operate on 110 v., single phase, 60 cycle (c.), a.c., obtained from a 440/110 volt control transformer. On the 250 v., direct current (d.c.), sink heater, the control system shall operate from the nominal 250 v. (175 to 355 v.) d.c. system.

3.3.3.2 Heaters.- Immersion type electric heaters shall be provided and installed in the heating reservoir or boiler for heating the unit. The heaters shall be readily replaceable, and shall be capable of raising the temperature of a full sink (16-inch level) of water from 130°F. to 180°F. in not more than 30 minutes when tested in accordance with 4.2.3.5. Heating units shall be suitable for operation on 440 v., three phase, 60 c., a.c. or on 250 v. nominal (175 to 355 v.) d.c., as specified (see 6.1).

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3.3.3.2.1 Cutout protection, with manual reset, shall be provided to prevent heater damage when the heater water reservoir is partially or completely empty of water.

3.3.3.3 Control box. - A suitable control box for installation above the sink, with which the heater is to be used, shall be provided. The control box shall contain the on-off switch, indicator lights, contactors, relays, and electrical connection blocks. The electrical cable connecting the control box with the sink heater shall be not less than 8 feet long.

3.3.3.4 Protective features. - The following protective features shall be provided:

(a) Automatic temperature control to limit the temperature of water to between 180°F. and 190°F. when tested in accordance with 4.2.3.5.
(b) A high temperature cutout, with manual reset, that will open and cut off power to the heaters and prevent damage when sink water temperature has reached 205°F. to 210°F., or when the sink is dry (without water). This feature may be combined with the cutout protection specified in 3.3.3.2.1, provided that both functions can be accomplished.
(c) A green signal light to indicate when the water in the sink is at, or above, a temperature of 180°F.
(d) A yellow signal light to indicate when the unit is energized.
(e) A red signal light to indicate that the sink or reservoir is dry, or that water temperature has exceeded 205°F.
(f) A durable on-off switch for energizing the unit. The switch shall break all legs of the circuit when in the off position.
(g) A warning plate adjacent to the on-off switch, warning against turning the unit on without water in the sink.

3.3.3.5 Thermostat controls. - The thermostat controls shall be accessible from the front of the unit for adjustment. Provision shall be made to discourage tampering with the control settings, by a separate panel over the controls and sealing, or locking the control settings with recessed set screws or sealing compounds.

3.3.4 Insulation. - Sinks shall be insulated, as necessary, to conserve heat and protect personnel from injury.

3.3.5 Cover. - Each sink shall be provided with a tight flush-fitting cover.

3.4 Construction. - Units shall be of welded construction.

3.5 Finish. - Exposed surfaces shall be given a number four finish.

3.6 Identification plate. - An identification plate of durable material shall be secured in a conspicuous place on each sink unit. Marking shall be of a method which will be permanent and legible. The plate shall indicate the following:

(a) Sink size (size I or II).
(b) Electrical characteristics.
(c) Stock number (when given in contract or order).
(d) Manufacturer's name and address.
(e) Date of manufacture.
(f) Contract number.

3.7 Manuals. - Manuals shall be of a commercial type, consisting of manufacturers' standard literature, illustrations or drawings to outline concisely the proper installation, operation, maintenance and parts identification of the sink units. The manual shall be suitably bound in a self-cover of durable quality. Two copies of the manual shall be provided with each sink unit. Two copies shall be submitted to the Bureau of Ships, via the Inspector of Naval Material, for assignment of NAVSHIPS number. Additional copies for stock shall be as specified (see 6.1).

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3.8 Repair parts.- The following repair parts shall be furnished with each sink unit unless otherwise specified (see 6.1):

One complete set of heating elements.
One magnetic contactor for each type and size used.
One set of contacts for each size and type of relay used.

3.9 Workmanship.- Workmanship, in general, shall conform to the following:

(a) Welds are cleaned and ground to present a uniform and smooth finish.
(b) Controls are suitably located and hazard-free to operator.
(c) All surfaces are free of burrs and sharp edges.
(d) Sink interior is free of obstructions and has smooth interior finish.
(e) Insulation jacket accomplishes purpose for which intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction.- Prior to the start of production and within 45 days after the date of award of the contract or order, one sink unit of each size under the contract or order shall be submitted to the U.S. Naval Supply Research and Development Facility, Bayonne, New Jersey, for tests of 4.2.3.1 through 4.2.3.7. The preproduction sample is defined as a production line model of the product which the manufacturer will subsequently produce in fulfillment of the contract. Approval of the preproduction sample by the U. S. Naval Research and Development Facility will not relieve the contractor of the responsibility of compliance with the requirements of this specification. Any change or deviations from the preproduction sample shall be submitted for the approval of the bureau or agency concerned. The approved preproduction sample will be used as a guide in the inspection of production units, and shall be presented for delivery as the last unit for delivery under the contract.

4.2.1 Prior to the submission of the preproduction sample for testing, the manufacturer shall subject the sink unit to the test of 4.2.3.4. Certification containing actual test data shall be furnished showing that the sample submitted for preproduction tests has successfully passed the test of 4.2.3.4.

4.2.2 When a supplier has previously supplied equipment on a recent contract or order, identical to that covered by this specification, which has passed the preproduction tests specified herein, the requirements for preproduction tests may be waived by the bureau or agency concerned, provided acceptable test data is furnished.

4.2.3 Preproduction tests.-

4.2.3.1 Vibration and impact tests.- The sink unit shall be tested without packaging or crating. The vibration test shall be performed prior to the impact test. The sink unit tested shall be capable of performing its principal functions after testing.

4.2.3.1.1 Vibration test.- The sink unit shall be placed on the table of a vibration tester. Fences, suitable for the sink unit being tested, may be attached to the test table. While the sink unit is heating and maintaining the temperature of its rated capacity of water, it shall be vibrated so that it leaves the table momentarily at some interval during the vibration cycle. The proper vibration frequency shall be such that a 1/16 inch thick shim may be inserted between one bottom edge of the sink unit and the table of the vibration machine. The sink unit shall be vibrated for 2 hours. The function of this test is to produce a measured oscillation upon the equipment to determine its ability to withstand dynamic forces due to vibration which may be encountered aboard ship. The sink unit shall function satisfactorily during this test.

4.2.3.1.2 Impact test.- An inclined impact testing device similar to the Conbur machine, or equal equipment producing equivalent results, and a shock recorder shall be used. The sink unit shall be

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fastened to the base so that sink unit will make no physical contact with the barrier of the impact machine. The assembly shall be placed on the dolly with the face or edge of the base which is to receive the impact projecting approximately 2 inches beyond the forward end of the dolly. The shock recorder shall be positioned on the machine to record the maximum shock received during the impact test. The dolly shall be drawn up the incline to a position which will produce an impact which has a vertical component equivalent to 10*g*.

4.2.3.2 Overall construction and performance. - The sink unit shall be tested and checked to determine compliance with the requirements of 3.3.1 through 3.5, and 3.9.

4.2.3.3 Power input. - Heating units shall not draw in excess of 5 percent of rated current.

4.2.3.4 Insulation. - The insulation resistance of power carrying electrical components shall be measured by a 500 v., d.c., insulation resistance tester. Insulation resistance of each component shall be not less than 10 megohms.

4.2.3.5 Time and temperature control. - Three thermocouples shall be located in the sink under test, one to be located at a random location near the back of the sink and approximately 12 to 14 inches above the sink bottom; one to be geometrically centered in the sink at a distance of 8 inches above the sink bottom; and one to be located at random near the front of the sink and approximately 2 to 4 inches above the sink bottom. The sink under test shall then be filled to a 16 inch depth with 130°F. water and the heating unit turned on. The maximum time required for the temperature of any one of the three thermocouples to reach 180°F. shall not exceed 30 minutes. The unit shall continue to be operated for an additional four hours after attaining 180°F. Temperature shall be maintained between 180°F. and 190°F. over this period of time.

4.2.3.6 Functional test. - With unit operating and sink water temperature between 180°F. and 190°F, the sink shall be subjected to the following test:

(a) Sufficient immersion baskets shall be loaded with ten 9-inch dinner plates to simulate actual dishwashing and sanitizing conditions, or means shall be provided to lower the temperature of the dinner plates (after being removed from the sanitizing water) from 180 to 190°F. to ambient prior to again inserting them in the 115 to 125°F. water. The test shall consist of immersing a loaded basket, with dinner plates at ambient temperature, into water having a temperature of 115 to 125° F. for two minutes, then removing the loaded basket and placing it in the heated sink for one minute. This cycle shall be repeated for a total of 30 times. Temperature measurements shall be made by three thermocouples fixed at random locations within the heated sink where they cannot be dislodged or damaged by the immersion basket. The average of the 30 minimum temperatures recorded during each of the 30 immersion cycles shall be not less than 180°F. The average of the 30 maximum temperatures recorded during each of the 30 immersion cycles shall not exceed 190°F.

4.2.3.7 Phase unbalance. Power unbalance between phases of a.c. heaters shall not exceed 10 percent.

4.3 Material certification. - The contractor shall certify in writing that the materials used in the construction of the sink units furnished conform to the material requirements outlined herein. Such certification shall document actual tests, examination or other verifiable quality data.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

5.1.1 Sanitizing sinks. -

5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the using activity and until early installation, and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during

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direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation, and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the supplier's commercial practice. The information shall include nomenclature, size, Federal stock number or manufacturer's part number, contract or order number, supplier's name, and destination.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation and packaging, packing, and marking shall be specified by the preparing activity (see 6.1).

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of sinks required (see 1.2).
(c) Electrical characteristics (see 3.3.3.2).
(d) Quantity of manuals required for stock (see 3.7).
(e) Repair parts required, if other than specified (see 3.8).
(f) Preservation, packaging, packing or marking requirements other than those required by 5.1 (see 5.2).

6.2 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - Ships
(Project 4510-N022Sh)

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